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TOPIC 2: MONOHYBRID CROSS, MENDEL'S LAWS & TEST CROSS

PU-II Principles of Inheritance and Variation • Topic-Wise Worksheet

Student Name: ____ **Roll / Reg. No:** ____________ **Date / Batch:** _________

SECTION A: Multiple Choice Questions (1 Mark Each)

1. The phenotypic and genotypic ratios in the F2 generation of a monohybrid cross are:

(A) 3:1 and 1:2:1 (B) 1:2:1 and 3:1
(C) 9:3:3:1 and 1:1:1:1 (D) 1:1 and 3:1

2. A test cross is performed by crossing an individual with an unknown dominant phenotype with a:

(A) Homozygous dominant parent (B) Heterozygous hybrid
(C) Homozygous recessive parent (D) Any parent

3. Which Mendelian law states that paired alleles segregate without blending during gametogenesis?

(A) Law of Dominance (B) Law of Segregation
(C) Law of Independent Assortment (D) Law of Polygenic Inheritance

4. In guinea pigs, crossing homozygous black (BB) with homozygous white (bb) produces all black F1 because:

(A) White is dominant (B) Black is dominant over white
(C) Incomplete dominance occurs (D) Co-dominance occurs

SECTION B: Fill in the Blanks (1 Mark Each)

5. A graphical representation developed by Reginald C. Punnett to calculate genotype probability is called a ....

6. The Mendelian genotypic ratio (1:2:1) can be mathematically derived from the binomial expression ....

7. A monohybrid test cross gives a phenotypic ratio of ....

8. The Law of Segregation is also famously referred to as the law of .... of gametes.

SECTION C: Very Short Answer Questions (2 Marks Each)

9. **[2M]** State Mendel's Law of Dominance and give one limitation/exception to it.

10. **[2M]** What is a test cross? State its biological significance in plant breeding.

SECTION D: Short Answer Questions (3 Marks Each)

11. **[3M]** Using a Punnett square, show how a test cross distinguishes between a homozygous tall (TT) and a heterozygous tall (Tt) pea plant.

12. **[3M]** Explain the monohybrid inheritance of coat colour in guinea pigs with a complete schematic cross up to F2 generation.

SECTION E: Long Answer / Essay Questions (5 Marks Each)

13. **[5M]** Describe Mendel's monohybrid cross for stem height in *Pisum sativum* from parental generation to F2 generation. State the phenotypic and genotypic ratios and state the Law of Segregation derived from it.

14. **[5M]** What is a test cross? Diagrammatically represent test crosses for both homozygous dominant and heterozygous dominant genotypes in violet/white flowers and explain the results.